14.7 Second-order partial derivatives

Second-order partial derivatives are defined for $z = f(x,y)$ as

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$$

$$f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

$$f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

→ Do problem 1

<u>Fact</u> If f_{xy} and f_{yx} are continuous, then $f_{xy} = f_{yx}$.

→ Do problem 2.